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C. Non-Descending Arrays

time limit per test: 2 seconds

memory limit per test: 256 megabytes

You are given two integer arrays a and b both of length n .

You can choose any subset of indices and swap the elements at those positions (i. e. make a swap(a_i , b_i) for each i in the subset). A subset of indices is considered **good** if, after the swapping, both arrays are sorted in non-descending order.

Your task is to calculate the number of good subsets. Since the answer can be large, print it modulo 998244353.

Input

The first line contains a single integer t ($1 \leq t \leq 500$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 100$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 1000$).

The third line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 1000$).

Additional constraint on the input: there is at least one good subset.

Output

For each test case, print a single integer — the number of good subsets, taken modulo 998244353.

Educational Codeforces Round 182 (Rated for Div. 2)

Contest is running

01:58:03

Contestant



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Language: Python 3.13.2 
 Almost always, if you send a solution on PyPy, it works much faster

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Example

input

Copy

```
3
3
2 1 4
1 3 2
1
4
4
5
2 3 3 4 4
1 1 3 5 6
```

output

Copy

```
2
2
8
```

Note

In the first example, there are 2 good subsets: {1, 3} and {2}.

In the second example, there are 2 good subsets: {1} and {}.

In the third example, there are 8 good subsets: {1, 2, 3, 4, 5}, {1, 2, 3}, {1, 2, 4, 5}, {1, 2}, {3, 4, 5}, {3}, {4, 5} and {}.

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